

Phase 4 Specification

OmniERP — Document Governance & Procure-to-Pay

4a Document Lifecycle & Approval Substrate

4b Purchasing, Receiving & Landed Cost

Ready for architectural design — 9–10 weeks

DOCUMENT	Phase 4 Specification
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AUTHOR	Principal Architecture & Product
AUDIENCE	Principal Systems Architect & coding assistant
STATUS	6 open questions (Q-19 to Q-24)

0. Phase Reconciliation

Your delivered phases do not map 1:1 onto SRS v2.0's numbering. Reconciling before specifying anything:

YOUR PHASE	DELIVERED	SRS V2.0 EQUIVALENT	STATUS
Phase 0	Foundation, outbox, ARQ, isolation, hash chain	P0	✓ Complete
Phase 1	Inventory depth + cost layer ledger + FIFO/WAC	P1a + P1b	✓ Complete
Phase 2	Omnichannel & sales engine, COGS double-hook, returns	P4 (sales portion)	✓ Complete, out of blueprint order
Phase 3	Accounting core, GL, trial balance, period close	P1c (partial)	⚠ Verify \$1
—	—	P1d — Approvals	✗ Not built
—	—	P2 — Purchases & Landed Cost	✗ Not built
—	—	P3 — ETA Compliance	✗ Not built. Now the critical path — see §0.2

0.1 Why Phase 4 Must Be Procure-to-Pay

Three arguments, in order of weight:

1. Your costing engine is running on one leg. You have built FIFO consumption, WAC proportional consumption, specific identification, and exact historical reversal on returns. All of that is the *outbound* half. The inbound half — the half that creates layers with real costs, real currencies, real FX rates, and real landed cost — does not exist. Every cost layer in your system today came from an opening balance or an adjustment. **FR-330 (retroactive landed cost), FR-331 (provisional costing), and FR-332 (backdated recalculation) are written but have never been exercised by a real business flow.** Those three are the hardest code in the entire product and they are currently unproven.

2. Approvals get exponentially more expensive with every module you ship. The approval state machine is *uniform across every document type* by design (SRS v2.0 §5.2). Retrofit cost scales with the number of document types already in production. Today that is sales invoices, credit notes, and journal entries — three retrofits. Build purchases first and it becomes seven. **The cheapest moment to build approvals is always now, and it gets worse every week.**

3. Landed cost is the requirement that wins Hossam. The importer persona is your entire secondary segment, and "I finally know my true per-unit cost" is the demo that closes them. It is unbuildable without procurement.

0.2 The Warning You Should Act On This Week

ETA e-invoicing is now your critical path and it has not started.

It is a 6-week subsystem with an external dependency (cloud HSM signing provider) whose procurement lead time you do not control. It is also legally load-bearing — the mandatory registration threshold sits at EGP 250,000 annual revenue under Resolution 281/2025, which captures essentially your entire target market.

Recommendation: begin signing-provider procurement in parallel with Phase 4a, week 1. Do not sequence it after Phase 4. If the third engineer has arrived, ETA integration is the ideal work package for them — it is cleanly separable from the costing and approval work and touches almost no shared code.

1. Verification Gates — Confirm Before Starting

Phase 4 assumes these are true. If any is false, it belongs in Phase 4 scope and changes the estimate.

#	ASSERTION	WHY PHASE 4 DEPENDS ON IT
V-1	<code>branch_id</code> exists on every transactional table and on <code>journal_lines</code> , not just document headers	Purchase documents and landed cost allocations are branch-dimensioned. Retrofitting the dimension after purchases exist means touching every table again
V-2	AR/AP control accounts exist and manual journal entry directly to them is blocked (FR-552)	The vendor bill subledger posts to AP control. If manual entry to AP is permitted, the subledger will diverge and you will not find out for months
V-3	Period close hard blocks on unposted documents and unbalanced entries	Phase 4 adds three new close gates (GRNI open items, unreconciled provisional costs, in-flight approvals). The harness must already exist
V-4	A rounding difference account exists and is used	You quantize to 4dp from 6dp unit costs. Over 100k lines, sub-piastre residue accumulates. "Zero COGS drift" on the cost-layer side does not imply zero drift on the GL side. Verify: post 10,000 lines with awkward unit costs and confirm the trial balance still nets to zero without manual intervention
V-5	Multi-currency: <code>journal_lines</code> carry both transaction-currency and base-currency amounts, stored not derived	Purchases are where FX gets genuinely hard (\$5.3). If base amounts are computed at read time, stop and fix that first
V-6	Auto-scaffolded accounts respect <code>is_system</code> protection and cannot be created inside a closed period	Your event consumers scaffold accounts automatically. That is convenient and slightly dangerous — an accountant who sees accounts appearing unbidden loses trust fast. Confirm scaffolding is bounded to a whitelist of known account roles, not open-ended
V-7	<code>stock_movements</code> immutability triggers are live and <code>cost_layers</code> permits only the four mutable columns	Goods receipt is the highest-volume writer to these tables

2. Phase 4 Architectural Goals

1. **One document lifecycle, not seven.** Every approvable document — existing and future — shares a single state machine, a single approval engine, and a single audit surface. No document type gets bespoke states.
2. **The approval engine becomes the universal safety rail.** It governs humans in Phase 4, integrations in Phase 6, and AI agents in Phase 8 with zero additional work. Design it as a general gate, not a purchasing feature.
3. **Purchasing never writes to the cost ledger directly.** It emits domain events; the costing context owns layer creation. Your bounded-context discipline holds — purchasing knows nothing about FIFO.

4. **Landed cost is a first-class allocation engine**, not a field on a bill. It must handle estimated-then-actual, multi-currency, multiple allocation bases, and application *after* goods have already been sold.
 5. **Quantity truth lives in one place**. PO ordered, GRN received, bill billed, and stock on-hand must reconcile continuously. Three-way match is an invariant, not a report.
 6. **Every FX event is recognised at the moment it occurs**. A purchase generates up to three distinct FX differences with three distinct accounting treatments (§5.3). Getting this wrong is silent and expensive.
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PHASE 4a — Document Lifecycle & Approval Substrate

Estimated: 4 weeks · Build first. Everything in 4b sits on top of it.

3a.1 Domain Entities

```

DocumentLifecycle      (mixin / protocol – not a table)
  state                draft|pending_approval|approved|posted|rejected
                      |cancelled|withdrawn|reversed|closed
  content_hash         sha256 of the approvable content (FR-1212)
  submitted_at, submitted_by
  approved_at, posted_at
  reversal_of_id

ApprovalRule
  id, document_type, is_active, version
  condition_field       total_amount|currency|branch|warehouse|contact
                      |item_category|discount_pct|margin_pct|payment_terms
  operator              gt|gte|lt|lte|eq|in|not_in
  condition_value       jsonb
  approver_type         specific_user|role|creator_manager|approval_group
  approver_ref          uuid
  sequence_no           int      -- equal values = parallel approval
  is_mandatory          bool
  effective_from, effective_to

ApprovalGroup / ApprovalGroupMember
  semantics             any_of | all_of

ApprovalRequest
  id, document_type, document_id, document_content_hash
  rule_id, rule_version
  sequence_no
  state                 pending|approved|rejected|withdrawn|escalated|expired
  requested_at, requested_by
  due_at
  resolved_at, resolved_by, resolution_comment
  delegated_from_user_id

ApprovalDecision        (immutable audit – FR-1211)
  id, approval_request_id
  decision               approve|reject
  decided_by, decided_at, ip_address, user_agent
  comment
  document_content_hash  -- what was actually approved

ApprovalDelegation
  delegator_id, delegate_id, document_types[], valid_from, valid_to, reason

MasterDataApproval      (FR-1217 – four-eyes on non-document changes)
  entity_type            contact_bank_details|item_price|credit_limit|supplier_bank
  entity_id
  proposed_change        jsonb
  state, requested_by, approved_by

```

3a.2 Functional Requirements

Restated from SRS v2.0 §5 with Phase 4 implementation notes.

ID	REQUIREMENT	IMPLEMENTATION NOTE
FR-1201	Uniform state machine across Sales Invoice, Credit Note, Journal Entry, and all Phase 4b documents	Implement as a mixin + service, not per-module logic

ID	REQUIREMENT	IMPLEMENTATION NOTE
FR-1202	Illegal transitions rejected at the database level	Trigger or CHECK on a (state_from, state_bated) transition table. Your Phase 1 immutability triggers are the pattern to follow
FR-1203	Declarative rule table. Conditions strictly bounded — no scripting, no expression language, no visual designer	This is the single most important scope discipline in Phase 4. See §5.5
FR-1204	Multiple matching rules all apply. Sequential by sequence_no , parallel when equal	
FR-1205	Approver resolution: user, role, creator's manager, or group with any-of/all-of	
FR-1206	Segregation of duties. Creator can never approve own document, regardless of role. System must refuse to create an unsatisfiable rule rather than deadlock documents	DB-level constraint. This is also what makes the AI agent safe in Phase 8
FR-1207	Amount thresholds evaluated in tenant base currency using the document's FX rate	A USD 1,100 PO must trip an "over EGP 50,000" rule
FR-1208	Delegation with date range, recorded on the approval record	
FR-1209	Escalation after N hours	Tier B — ARQ scheduled job
FR-1210	Rejection requires mandatory reason; returns to draft with reason visible	
FR-1211	Immutable approval audit: who, when, IP, device, comment, rule version, document content hash	Append-only, same trigger pattern as cost_consumptions
FR-1212	Document content hash. Modifying an approved document in draft voids the prior approval	Hash the <i>approvable</i> fields only — lines, amounts, contact, dates. Not updated_at , not internal notes
FR-1213	Unified approval inbox across all document types, with filters and bulk approve	
FR-1214	Mobile approval flow (PWA) — push → summary card → approve/reject with comment	Approvers are owners and managers; they are not at a desk. If approving is inconvenient, documents stall and the feature becomes a liability
FR-1216	Emergency override by Owner, typed reason, high-visibility audit event, notification to all managers	Every real business needs this. Omitting it means users route around the system entirely
FR-1217	Master-data four-eyes: selling price below margin floor, credit limit increase, supplier bank detail change	Supplier bank detail change is the highest-value fraud control in the product
FR-1218	Rules versioned; changing a rule does not affect in-flight approvals	

3a.3 The Retrofit — Existing Documents

You have three document types in production without lifecycle states. This is the awkward part of 4a and it needs a plan, not improvisation.

Sequence:

1. **Expand.** Add `state`, `content_hash`, and lifecycle timestamps to `sales_invoices`, `credit_notes`, `journal_entries` as nullable columns. No behaviour change.
2. **Backfill.** Every existing posted document gets `state = 'posted'`. Every draft gets `state = 'draft'`. Run as a per-tenant ARQ job through your migration orchestrator.
3. **Migrate.** Deploy code that reads and writes `state` but tolerates NULL.
4. **Contract.** Make `state` NOT NULL, install the transition trigger, enable rule evaluation.
5. **Default-off. A tenant with zero configured approval rules must see no approval UI and experience no behaviour change** — documents transition `draft` → `posted` directly. Approvals are opt-in per tenant.

Point 5 is critical for adoption. Most of your SMB tenants will never configure a rule, and they must not pay a UX tax for a feature they don't use.

3a.4 Events

```
Published:  document.submitted_for_approval
            document.approved
            document.rejected
            document.approval_escalated
            masterdata.change_requested
            masterdata.change_approved

Consumed:   (none in 4a – 4b and later phases consume these)
```

3a.5 Exit Criteria — 4a

1. ☒ A user cannot approve their own document via any path — API, bulk action, delegation chain, or direct DB write
 2. ☒ Altering an approved document in `draft` and resubmitting voids the prior approval (hash mismatch test)
 3. ☒ A USD-denominated document trips an EGP-denominated threshold rule correctly
 4. ☒ A tenant with no rules configured sees zero behavioural change vs. today
 5. ☒ Attempting an illegal state transition raises at the database level, not just the service layer
 6. ☒ Approval audit rows cannot be updated or deleted
 7. ☒ Mobile approve/reject round-trip works on a real phone over 3G
 8. ☒ Creating an unsatisfiable rule (single-user tenant, sequential self-approval) is rejected at rule-creation time
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PHASE 4b — Purchasing, Receiving & Landed Cost

Estimated: 5–6 weeks · The heart of Phase 4.

4b.1 Domain Entities

```

PurchaseRequisition / RequisitionLine
  requested_by, needed_by, justification, branch_id
  → approval gated

PurchaseOrder / PurchaseOrderLine
  supplier_id, po_number, order_date, expected_date
  currency, fx_rate, branch_id, warehouse_id
  payment_terms, incoterm
  line: item_id, variant_id, qty_ordered, qty_received, qty_billed,
        unit_price, discount, tax_code_id, expected_landed_unit_cost
  state: draft|pending_approval|approved|partially_received|received
        |partially_billed|closed|cancelled

GoodsReceipt / GoodsReceiptLine          -- the ONLY creator of inbound stock
  po_id (nullable – direct receipts allowed)
  supplier_id, receipt_date, warehouse_id, branch_id
  supplier_delivery_ref
  line: po_line_id, item_id, qty_received, qty_rejected,
        batch_id, serial_ids[], unit_cost_estimated
  → emits purchase.goods_received

VendorBill / VendorBillLine
  supplier_id, bill_number, bill_date, due_date
  currency, fx_rate, branch_id
  match_state: unmatched|matched|variance_within_tolerance|variance_blocked
  line: po_line_id, grn_line_id, qty_billed, unit_price, tax_code_id
  → emits purchase.bill_posted

ThreeWayMatchResult
  bill_line_id, po_line_id, grn_line_id
  qty_variance, price_variance, qty_variance_pct, price_variance_pct
  within_tolerance: bool, tolerance_rule_id
  resolution: auto_approved|routed_for_approval|blocked

MatchToleranceRule
  scope: tenant|supplier|item_category
  qty_tolerance_pct, qty_tolerance_abs
  price_tolerance_pct, price_tolerance_abs
  action_on_breach: warn|route_approval|block

ImportShipment                          -- the importer's container
  shipment_ref, supplier_id(s), po_ids[]
  stage: draft|lc_opened|paid|shipped|in_customs|cleared|received|closed
  departure_date, arrival_date, clearance_date
  bl_number, container_no, incoterm
  estimated_total_landed, actual_total_landed

LandedCostLine
  shipment_id
  cost_type: freight|insurance|customs_duty|clearance|inland_transport
            |bank_charges|demurrage|other
  supplier_id, currency, fx_rate, amount, tax_code_id
  allocation_basis: value|weight|volume|quantity|manual
  is_estimated: bool, actual_of_estimate_id

LandedCostAllocation                    -- immutable
  landed_cost_line_id
  target_grn_line_id
  allocated_amount
  cost_layer_id                        -- resolved after layer creation
  is_retroactive: bool                -- applied after goods issued
  cogs_trueup_journal_entry_id

GRNIItem                                -- goods received not invoiced
  grn_line_id, accrued_amount, accrued_at
  cleared_by_bill_line_id, cleared_at

```

```

SupplierPayment / PaymentAllocation
  treasury_id, payment_date, currency, fx_rate, amount
  allocation: bill_id, allocated_amount
  fx_gain_loss_amount

```

4b.2 Functional Requirements

ID	REQUIREMENT	PRI
Procurement flow		
FR-520	Requisition → PO → Goods Receipt → Vendor Bill → Payment, with approval gates on requisition and PO via the 4a engine	A
FR-521	Three-way match with configurable tolerances (pct and absolute, scoped tenant/supplier/category). Within tolerance → auto-approve. Outside → route for approval or block per rule	A
FR-522	Partial receipt and partial billing against a PO, with open-quantity tracking. <code>qty_ordered ≥ qty_received ≥ qty_billed</code> must hold as an invariant, not a convention	A
FR-527	GRNI accrual. Goods received but not billed accrue to a Goods-Received-Not-Invoiced liability account, cleared on bill posting. Aged GRNI report	A
FR-529	Purchase returns / debit notes with serial and batch validation, reversing the exact cost layers consumed	A
FR-530	Supplier performance: on-time delivery rate, price variance trend, rejection rate	B
FR-528	Supplier price agreements with validity dates and automatic price application	B
Import & landed cost		
FR-523	Import shipment lifecycle: PO(s) → LC/advance payment → shipped → in customs → cleared → received. Each stage with its own dates and GL treatment	A
FR-524	Unlimited landed cost lines , each with own supplier, currency, FX rate, and tax treatment. Allocation basis selectable per cost line: value, weight, volume, quantity, or manual	A
FR-525	Estimated landed cost applied at receipt; actual applied when the real invoice arrives , with variance posting. Retroactive application per FR-330 when goods are already sold	A
FR-526	Landed cost variance report — estimated vs actual, per shipment, per cost type	A
FR-531	Allocation must be exact. Allocated amounts across targets must sum to the cost line amount to the piastre. Rounding residue goes to a designated target (largest by basis), never lost	A
FR-532	Landed cost allocation preview before commit — the merchant sees per-unit impact before posting	A
Payments		
FR-505	Supplier aging, payable schedule, payment run	A
FR-533	FX gain/loss recognised on settlement, distinct from revaluation	A

ID	REQUIREMENT	PRI
FR-534	Advance payments / supplier prepayments with application to future bills	A
FR-594	Post-dated cheque issuance to suppliers with maturity tracking	A

4b.3 Bounded Context Interfaces

Purchasing must not import inventory or costing models. It publishes; the costing context reacts.

```

purchasing → publishes:
  purchase.order_approved      {po_id, supplier_id, lines[]}
  purchase.goods_received      {grn_id, warehouse_id, branch_id,
                                lines[{item_id, variant_id, qty,
                                      unit_cost_base, currency, fx_rate,
                                      batch_id?, serial_ids[]?}],
                                occurred_at}
  purchase.bill_posted         {bill_id, supplier_id, total_base, tax, lines[]}
  purchase landed_cost_allocated {shipment_id, allocations[{grn_line_id,
                                                            amount_base}]}
  purchase.return_posted       {debit_note_id, lines[]}
  purchase.payment_made         {payment_id, allocations[], fx_gain_loss}

costing → consumes:
  purchase.goods_received      → creates CostLayer per line (idempotent on event_id)
  purchase landed_cost_allocated → raises unit_cost_current on open layers;
                                emits COGS true-up for consumed portions (FR-330)
  purchase.return_posted       → reverses layers at exact historical cost

accounting → consumes:
  purchase.goods_received      → Dr Inventory  Cr GRNI
  purchase.bill_posted         → Dr GRNI, Dr Tax  Cr AP
  purchase landed_cost_allocated → Dr Inventory  Cr Landed Cost Clearing
                                (+ COGS true-up if retroactive)
  purchase.payment_made         → Dr AP  Cr Treasury  ± FX Gain/Loss

```

Interface contract requirement: `purchase.goods_received` carries `unit_cost_base` already converted to tenant base currency. The costing context must never perform FX conversion — that is purchasing's responsibility, and it keeps the cost ledger's currency semantics (all costs in base currency) intact.

4b.4 GL Posting Map

Give this table to the coding assistant verbatim; it removes most of the ambiguity.

EVENT	DEBIT	CREDIT
Goods received	Inventory (at estimated landed cost)	GRNI Accrual
Vendor bill posted (matched)	GRNI Accrual, Input VAT	Accounts Payable
Bill price variance (within tolerance)	Purchase Price Variance	GRNI Accrual
Landed cost line recorded	Landed Cost Clearing	Accounts Payable / Treasury

EVENT	DEBIT	CREDIT
Landed cost allocated to open layers	Inventory	Landed Cost Clearing
Landed cost allocated to consumed qty	COGS (true-up, current period)	Landed Cost Clearing
Purchase return	GRNI Accrual / Accounts Payable	Inventory (at exact historical layer cost)
Supplier payment	Accounts Payable	Treasury
FX difference on settlement	FX Loss <i>(or credit FX Gain)</i>	Accounts Payable
Withholding tax on payment	Accounts Payable	WHT Payable
Advance to supplier	Supplier Advances (asset)	Treasury

5. The Hard Problems in Phase 4

Each of these has broken a real ERP. Design for them explicitly.

5.1 Three-Way Match Across Partial Documents

One PO for 100 units. Three receipts of 40/35/25. Two bills covering 60 and 40 — but the second bill spans receipt 2 and receipt 3, and the supplier billed a different unit price on the second.

Requirement: matching is at **line-fragment granularity**, not document granularity. A `ThreeWayMatchResult` links a bill line to *one or more* GRN lines with a quantity split. Attempting document-level matching produces a system that works in the demo and fails in week two.

Invariant to enforce continuously: for every PO line, `qty_billed ≤ qty_received ≤ qty_ordered` (allowing an over-receipt tolerance if configured). Add to the nightly reconciliation job.

5.2 Retroactive Landed Cost Across a Closed Period

Goods received in January, sold in January, February closed. The freight invoice arrives in March.

You built FR-330 in Phase 1. Phase 4 is where it finally runs. The rule, restated because it is the one people get wrong: **increase `unit_cost_current` on the remaining layer quantity; for the already-consumed portion, post a COGS true-up dated in the current open period, referencing the original consumptions via `adjustment_of_id`.** Never edit the January entry. Never restate a closed period. The January margin report is annotated as adjusted, with a link.

Test that must exist: receive → sell everything → close the period → apply landed cost → assert the original journal entry is byte-identical, a new true-up entry exists in the open period, and stock valuation still reconciles to the inventory GL account.

5.3 The Three FX Events

A single import generates up to three distinct FX differences, each treated differently. Most implementations collapse them and misstate cost.

MOMENT	RATE	TREATMENT
PO raised	Rate A	No accounting. A PO is a commitment, not a transaction
Goods received	Rate B	Sets inventory cost. This rate, and only this rate, determines the cost layer's <code>unit_cost_original</code> . It is never revisited
Bill posted	Rate C	Difference between B and C on the <i>inventory</i> portion is a purchase price variance , not an FX gain — the goods are already valued
Payment settled	Rate D	Difference between C and D is a realised FX gain/loss on the AP liability
Period end, unpaid	Rate E	Unrealised FX revaluation on the open AP balance, reversed next period

The rule to encode: inventory cost is fixed at receipt (rate B) and never moves for FX reasons. Everything after that is a P&L event on the liability, not a cost adjustment. Getting this backwards means your inventory value drifts with the exchange rate, which is both wrong and unauditible.

5.4 Allocation Rounding

Allocating EGP 10,000 freight across 7 items by value will not divide evenly. Naive per-item rounding loses or gains piastres, and the inventory GL will not match the stock valuation — which trips your own invariant job.

Requirement (FR-531): allocate using largest-remainder, and assign the residue to the single largest target by basis. Assert `SUM(allocations) == cost_line.amount` as a hard constraint before commit. Not a warning. A constraint.

5.5 Approval Rule Scope Discipline

Someone will ask for a visual workflow designer, or conditions with AND/OR nesting, or a scripting hook. **Refuse all three.**

The bounded rule table in FR-1203 covers approximately 95% of real SMB approval needs. A general workflow engine is a six-month project that 4% of tenants will use, that will produce your worst bugs, and that will make every document state transition slow and unpredictable. If a tenant genuinely needs graph-shaped approvals, that is a Phase 9 conversation with a paying enterprise customer, not a Phase 4 feature.

This is the single most important scope discipline in Phase 4. Write it on the wall.

6. Phase 4 Exit Criteria

Approvals (4a) 1. ☒ All eight 4a criteria in §3a.5

Procure-to-Pay (4b) 2. ☒ Full flow: requisition → approved PO → 3 partial receipts → 2 bills spanning receipts → payment, with correct GL at every step 3. ☒ `qty_billed ≤ qty_received ≤ qty_ordered` holds under a concurrent load test 4. ☒ GRNI account nets to zero when all receipts are fully billed 5. ☒ Three-way match auto-approves within tolerance and blocks outside it 6. ☒ **Retroactive landed cost across a closed period** posts a true-up in the open period, leaves the closed period byte-identical, and stock valuation reconciles to the GL to the piastre 7. ☒ Landed cost allocation across 7 items by value sums exactly to the cost line amount 8. ☒ The three FX events (§5.3) each post

to the correct account with the correct rate 9. ☒ Import shipment lifecycle progresses through all stages with correct GL at each 10. ☒ Purchase return reverses at exact historical layer cost — zero drift, matching your sales-return standard 11. ☒ Nightly invariant job extended with: GRNI reconciliation, PO quantity invariant, landed cost allocation exactness, provisional cost queue depth 12. ☒ Period close blocked by: open GRNI beyond threshold, in-flight approvals, unreconciled provisional costs

7. Resequenced Roadmap From Here

Given your renumbering, the remaining path:

PHASE	CONTENT	EST.	NOTE
4	Approvals + Procure-to-Pay + Landed Cost	9–10 wks	This document
5	ETA Compliance	6 wks	Start provider procurement NOW, run in parallel with 4
6	POS, UX transformation, design system, command palette, onboarding, Excel import, PWA	8 wks	The merchant-facing layer
7	Trust Network hardening, Bosta/Mylerz, COD settlement reconciliation	5 wks	The moat + the primary persona's biggest pain
8	Launch readiness: pen test, load test, DR drill, legal, billing, pilot	5 wks	
9	Tier B: consolidation, budgets, fixed assets, quality/quarantine, bins, kits	10 wks	Post-launch
10	AI subsystem (Blueprint I)	23 wks	Post-launch, revenue-funded

~33 weeks to launch from today, assuming ETA runs parallel to Phase 4. If ETA runs sequentially instead, add 6 weeks.

8. Open Questions

#	QUESTION	BLOCKS
Q-19	Confirm V-1 through V-7 in §1. Any "no" changes the Phase 4 estimate	Scoping
Q-20	Has ETA signing-provider procurement started? If not, this week	Phase 5, and therefore launch
Q-21	Did Phase 3 include the branch dimension on <code>journal_lines</code> , or headers only?	4b data model
Q-22	Third engineer — hired? If yes, they take ETA in parallel starting now	Timeline

#	QUESTION	BLOCKS
Q-23	Do your importer design partners exist yet? Landed cost allocation basis preferences (value vs weight vs volume) should be validated against two real shipments before you build the UI	4b design
Q-24	Should requisitions be in V1 at all, or is PO-as-entry-point sufficient for SMB? Requisitions add a document type and an approval layer that a 6-person trader may never use	4b scope

Q-20 is the one that matters. Everything else is a design detail; that one is on the critical path to launch.